Direct & Alternating Current

Exam code: 0625 & 0972

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Written by: Katie M
Reviewed by: Caroline Carroll
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Direct & alternating current

Help me understand

- There are **two** types of current
  - direct current (d.c.)
  - alternating current (a.c.)

Direct current

- A direct current (d.c.) is defined as

A steady current, constantly flowing in the same direction in a circuit, from positive to negative

- The potential difference across a cell in a d.c. circuit travels in **one direction only**
  - The current travels from the positive terminal to the negative terminal
- A d.c. power supply has a **fixed** positive terminal and a fixed negative terminal
- Electric **cells**, or **batteries**, produce direct current (d.c.)

Alternating current

- An alternating current (a.c.) is defined as

A current that continuously changes its direction, going back and forth around a circuit

- An alternating current power supply has two identical terminals that **change** from positive to negative and back

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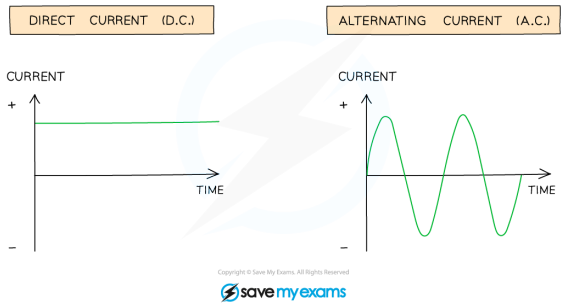
Got it

- Therefore, the current **changes direction** as the polarity of the terminals changes

- The **frequency** of an alternating current is the number of times the current changes direction back and forth each second
  - In the UK, **mains electricity** is an **alternating** current with a frequency of 50 Hz and a potential difference of around 230 V

Graphs of direct current and alternating current



Two graphs showing the variation of current with time for alternating current and direct current

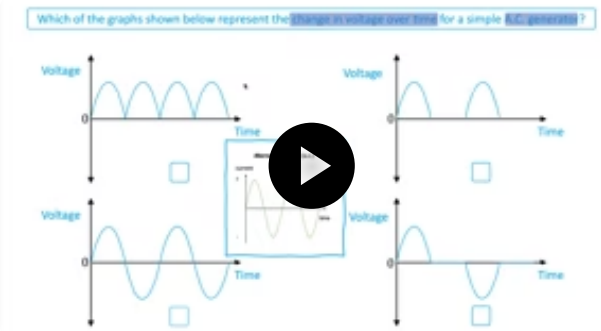
Comparing direct and alternating current

- The following table summarises the differences between d.c. and a.c.

Direct current vs. alternating current table

Direct current (d.c.)	Alternating current (a.c.)
continuous and in one direction	constantly changing direction
produced by cells and batteries	produced by electrical generators i.e. mains electricity
involves a positive and negative terminal	involves two identical terminals

Examiner Tips and Tricks



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